Week 2 Goal

In Week 2, your focus is NOT dashboard creation.
You must first understand, clean, and structure the data like a Business Analyst.
If your data is wrong → your insights will be wrong.

Week 2 is about **understanding the data deeply** before doing any charts, dashboards, or AI prompts.

If data is wrong → dashboard is wrong
If business logic is unclear → insights are meaningless

So Week 2 focuses on:

- What data we have
- What the business expects
- How to clean & structure data

1.Sales & Business Data Cleaning & Visualization

- **Domain:** Retail / FMCG / Corporate Sales etc..

Step 1 – Dataset Understanding

Explain columns :

- Sales Date → When sale happened
- Region → Location of sales
- Product → Item sold
- Revenue → Sales amount
- Profit → Earnings after cost
- Quantity → Units sold

Kindly ensure that you ask at least 7 to 8 questions, as outlined below.

- Which region generates the highest sales revenue?
- Which product contributes the highest profit margin?
- Which product category has the highest demand?
- Which region shows the fastest sales growth?
- Which products are generating losses?
- Which customer segment contributes the most revenue

- What is the sales trend over time (monthly/quarterly)?
- Which region has the highest return or cancellation rate?

<u>Step 2 – Data Cleaning</u>

- **Remove duplicate sales entries**
- **Fix date formats (DD/MM/YYYY)**
- **Convert currency symbols**
- **Handle blank profit values**

  - Missing values → blank sales amount, region missing
  - Duplicates → same invoice entered twice
  - Formatting → dates, currency symbols

<u>Step 3: Business KPIs Explanation</u>

- **Total Sales Revenue** → Overall company growth indicator
- **Total Profit** → Net earnings after deducting costs
- **Profit Margin (%)** → Business sustainability and pricing efficiency
- **Region-wise Sales** → Market performance comparison
- **Top Performing Region** → Strongest revenue-generating location
- **Best-Selling Product (by Quantity)** → Highest demand item
- **Most Profitable Product** → Highest contribution to profit
- **Loss-Making Products** → Items reducing overall profitability
- **Sales Growth Rate (Monthly/Quarterly)** → Business expansion trend
- **Average Revenue per Sale** → Sales efficiency indicator
- **Customer Segment Revenue Contribution** → Target market strength
- **Return / Cancellation Rate** → Revenue leakage measure
- **Revenue per Unit Sold** → Pricing effectiveness
- **Profit per Region** → Regional profitability comparison
- **Sales Contribution by Product Category** → Portfolio performance

<u>Step 4: KPI list</u>

Total Sales

Total Profit

Profit Margin (%)

Top Region (by Sales)

Best Product (by Sales or Profit)

Region-wise Sales

Monthly Sales Trend

Loss-Making Products

Average Revenue per Order

Sales Growth Rate

A manager never asks show me Excel rows, they ask Which region is performing best?

Step 5: Business Insight:

Sales KPIs help management understand where money is coming from, where it is leaking, and which products or regions drive profitability. Clean data ensures accurate strategic decisions.

2. Employee Payroll Data Cleaning & Financial Analysis

Domain: HR / Finance etc..

Real-World Context

Payroll data is sensitive and financial, errors can cause:

- Salary disputes
- Compliance issues
- Audit failures

Step 1: Dataset Explanation

Columns include:

- Employee ID
- Department
- Basic Salary
- Bonus
- Deductions
- Net Salary

Dataset Understanding(finanace)

- Employee ID → must be unique

- Gross Salary → before deductions
- Net Salary → final payout

Kindly ensure that you ask at least 7 to 8 questions, as outlined below.

Which department has the highest total payroll cost?

Which department has the highest average salary?

Are there any employees receiving unusually high or low salaries?

Which department contributes the highest tax deductions?

Are there duplicate employee records affecting payroll?

What percentage of total payroll goes toward bonuses?

Is the net salary correctly calculated for all employees?

Which department has the highest deduction ratio?

Step 2: Cleaning Tasks

- Remove duplicate employees
- Check incorrect salary values
- Standardize department names
- Validate deduction formulas

Cleaning Logic(finanace)

- Duplicate employees → salary paid twice
- Incorrect tax calculation
- Missing department mapping

Step 3: Business Analysis KPIs

Total Payroll Cost → overall company salary expenditure

Average Salary per Department → departmental pay benchmarking

Highest Paid Department → cost concentration analysis

Total Deductions → tax and statutory compliance tracking

Bonus Percentage of Payroll → incentive structure analysis

Net vs Gross Salary Ratio → deduction impact measurement

Payroll Growth Rate → workforce expansion indicator

Overtime Cost Percentage → operational efficiency tracking

Salary Variance Across Departments → pay equity analysis

Duplicate Payroll Entries → financial risk control

Step 4: KPI list

Total Deductions

Net Payroll Cost

Average Salary per Department

Highest Paid Department

Bonus Percentage of Payroll

Salary Variance Across Departments

Overtime Cost Percentage

Payroll Growth Rate

Deduction Ratio (Deductions/Gross Salary)

Duplicate Payroll Records Count

Tax Contribution by Department

Step 5: Business Insight

Finance teams use payroll KPIs to manage cost, ensure compliance, and plan hiring strategically.

3. Banking Customer Account Dashboard

Domain: Banking / Financial Services

Step 1: Dataset Understanding

Columns:

- **Customer Name**
- **Account Type**
- **Balance**
- **Transactions**
- **Loan Status**

Kindly ensure that you ask at least 7 to 8 questions, as outlined below.

Which account type holds the highest total balance?

Which customer segment maintains the highest average balance?

What percentage of accounts are inactive?

Which customers have high loan exposure?

Are there accounts with negative or suspicious balances?

Which account type generates the highest transaction activity?

What is the bank's total loan exposure?

Which customers are high-value but low-activity?

Step 2: Cleaning Activities

- **Remove inactive accounts**
- **Fix negative balances**
- **Categorize account types**

Step 3: Business KPIs

Total Deposits → bank liquidity strength

Total Loan Exposure → credit risk measurement

Average Customer Balance → customer financial value

Active vs Inactive Account % → engagement health

Loan-to-Deposit Ratio → lending efficiency

High-Risk Customer % → credit risk profiling

Transaction Frequency → customer activity level

Non-Performing Loan % → repayment risk indicator

Average Loan Amount → borrowing trend analysis

Customer Retention Rate → long-term relationship value

Step 4: KPI List

Inventory Turnover Ratio

Stock-Out Rate

Order Fulfillment Rate

Backorder Percentage

Average Delivery Time

Delayed Shipment %

Return Rate

Dead Stock Value

Top-Selling SKU

Category-wise Sales Contribution

Inventory Holding Cost (if available)

Order Cancellation Rate

Step 5: Business Insight:

Banking KPIs help identify high-value customers, risky borrowers, and inactive accounts. Monitoring loan exposure and deposit trends supports liquidity management and credit risk reduction. Clean customer data enables banks to make informed lending decisions and strengthen financial stability.

4. Online Store Inventory & Fulfilment

Domain: eCommerce / Supply Chain

Step 1: Dataset Explanation

- Product ID
- SKU
- Stock Level
- Orders
- Delivery Status

Kindly ensure that you ask at least 7 to 8 questions, as outlined below.

Which products are out of stock most frequently?

Which category generates the highest number of orders?

What is the fulfillment rate percentage?

Which products have delayed deliveries?

Which SKU has the highest inventory turnover?

Are there duplicate or cancelled orders affecting stock levels?

Which products have negative stock values?

Which category is most profitable (if price is available)?

Step 2: Cleaning Work

- Remove duplicate orders
- Fix stock mismatches
- Standardize product categories

Step 3: Business KPIs

- Inventory Turnover Ratio → stock movement efficiency
- Stock-Out Rate → lost sales opportunity measure
- Order Fulfillment Rate → delivery performance indicator
- Backorder Percentage → demand planning gap
- Average Delivery Time → logistics efficiency
- Delayed Shipment % → service quality measure
- Return Rate → product satisfaction indicator
- Top-Selling SKU → revenue driver identification
- Dead Stock Value → capital blockage analysis
- Category-wise Sales Contribution → product performance tracking

Step 4: KPI List

Inventory Turnover Ratio

Stock-Out Rate

Order Fulfillment Rate

Backorder Percentage

Average Delivery Time

Delayed Shipment %

Return Rate

Dead Stock Value

Top-Selling SKU

Category-wise Sales Contribution

Inventory Holding Cost

Order Cancellation Rate

Step 5: Business Insight:

Inventory insights reveal stock efficiency, demand trends, and fulfillment performance. High stock-out rates indicate lost sales opportunities, while dead stock ties up capital. Efficient inventory turnover improves profitability and customer satisfaction through faster and reliable deliveries.

5. Student Academic Performance Analysis

Domain: Education

Step 1: Dataset Columns

- Student Name
- Subject
- Marks
- Attendance

Kindly ensure that you ask at least 7 to 8 questions, as outlined below.

Which subject has the highest class average?

Which subject has the lowest pass percentage?

Who is the overall topper in the class?

Does attendance impact student performance?

What is the overall pass/fail percentage?

Which grade range has the highest number of students?

Are there students with high attendance but low marks?

Which subject appears most challenging based on results?

Step 2: Cleaning Tasks

- Fill missing marks
- Remove absent duplicates

- Standardize subject names

Step 3: Business KPIs

- **Class Average Score** → teaching effectiveness
- **Subject-wise Average** → curriculum difficulty analysis
- **Pass Percentage** → academic success rate
- **Attendance Rate** → student engagement level
- **Top Performer Score** → excellence benchmark
- **Fail Rate by Subject** → improvement focus area
- **Grade Distribution** → performance spread analysis
- **Attendance vs Marks Correlation** → behavioral impact study
- **Improvement Rate Over Time** → learning progress indicator
- **Low-Performing Student %** → intervention requirement

Step 4: KPI List

- Class Average Score
- Subject-wise Average
- Pass Percentage
- Fail Percentage
- Grade Distribution
- Top Performer Score
- Attendance Rate
- Attendance vs Marks Correlation
- Low-Performing Student %
- Improvement Rate Over Exams
- Subject Difficulty Index
- High Distinction %

Step 5: Business Insight

Academic data helps institutions identify subject difficulty, student engagement levels, and performance gaps. Correlating attendance with marks enables early intervention for struggling students. Data-driven decisions improve curriculum planning and overall learning outcomes.

6. BI Data Summarization using AI Prompts

Domain: Business Intelligence / AI

Step 1: Dataset Understanding

- Identify business domain (Sales / Finance / HR / Banking etc.)
- Understand each column meaning
    - Revenue → income generated
    - Cost → expenses
    - Profit → earnings after cost
    - Region → location performance
    - Date → trend analysis
- Identify KPI-related columns

Goal: Understand what story the data can tell.

Step 2: Data Cleaning & Preparation

- Remove duplicate rows
- Handle missing values (blank revenue, cost, etc.)
- Standardize date formats
- Convert currency symbols into numeric format
- Remove irrelevant columns
- Validate calculated columns (Profit = Revenue – Cost)

Important: AI gives wrong summaries if the data is incorrect.

Step 3: Business KPIs

- Total Revenue
- Total Profit
- Profit Margin
- Region-wise Sales
- Growth Rate
- Loss-Making Segments
- Top Performing Product/Region

AI works better when KPIs are already structured.

Step 4: Prompt Writing

Structured prompts

Basic Prompt Examples:

- Summarize overall sales performance.
- Identify the top-performing region.
- Highlight loss-making products.

Advanced Prompt Examples:

- Act as a Business Analyst. Provide executive-level insights from the dataset.
- Identify key risks and provide 3 recommendations.
- Summarize monthly growth trends in under 150 words.

Step 5: Compare AI Output with Actual KPIs

- Validate AI insights with calculated KPIs
- Identify incorrect AI assumptions
- Refine prompts if needed
- Ensure conclusions are data-backed

Step 6: Insight Generation

- Executive Summary (5–8 lines)
- Key Risks Identified
- Growth Opportunities
- Strategic Recommendations

Step 7: Create Prompt Library

- At least 10 well-structured prompts
- Basic prompts
- Analytical prompts
- Executive summary prompts
- Risk-identification prompts

Step 8: Business Insight:

- AI-driven summaries convert raw data into executive-level insights. Identifying trends, anomalies, and risks enables faster strategic decisions. Structured data and well-designed prompts enhance the accuracy and reliability of AI-generated business recommendations.

7. Content Creation & Marketing Copy Generation

Domain: Digital Marketing/Advertising

Step 1: Dataset Understanding

- **Audience Type** → Who is the target? (Students, Working Professionals, Parents, etc.)
- **Product Category** → What is being promoted?
- **Campaign Goal** → Awareness / Leads / Sales
- **Platform** → Instagram / LinkedIn / Email / Website
- **Keywords** → SEO & engagement drivers

Step 2: Data Cleaning

- Remove duplicate keywords
- Standardize tone (formal / friendly / premium)
- Categorize audience segments
- Remove irrelevant content themes
- Fix formatting inconsistencies

Step 3: Define Marketing KPIs

- Engagement Rate
- Click-Through Rate (CTR)
- Conversion Rate
- Cost per Lead (CPL)
- Return on Ad Spend (ROAS)
- Content Reach
- Email Open Rate
- CTA Effectiveness

Step 4: Prompt Writing Framework

Prompt Formula:

Audience + Goal + Tone + Platform + Word Limit

Example Prompts:

- Write an Instagram ad copy targeting working women promoting an AI course in a motivational tone under 100 words.

- Create a professional email campaign for MBA students inviting them to a data analytics webinar.
- Generate a product description for an e-commerce skincare brand in a premium tone.
- Write 5 high-converting CTAs for a fintech app.

Step 5: Content Evaluation

- Clarity
- Relevance to audience
- Strong CTA
- Emotional trigger
- Brand consistency

Step 6: Generate Marketing Assets

- Instagram Ad Copy
- LinkedIn Post
- Email Campaign Draft
- Product Description
- 5 CTA Variations
- 5 Headline Options

Step 7: Insight & Strategy Summary

- Which tone works best?
- Which platform needs shorter copy?
- Which audience responds to emotional vs logical messaging?
- How AI improved speed & productivity?

8. Hospital Patient Records

Domain: Healthcare

Step1: Dataset Columns

- Patient ID
- Age
- Disease

- Admission Date
- Department

Step 2: Cleaning Tasks

- Remove duplicate patients
- Standardize disease names
- Fix date errors

Step 3: Business KPIs & Explanation

- **Total Patient Admissions** → hospital demand level
- **Bed Occupancy Rate** → capacity utilization
- **Average Length of Stay** → treatment efficiency
- **Department-wise Patient Load** → workload distribution
- **Readmission Rate** → treatment quality indicator
- **Disease Frequency Trend** → public health monitoring
- **Age Group Distribution** → demographic insights
- **Emergency Admission %** → critical care demand
- **Discharge Rate** → recovery efficiency
- **Data Error Rate** → patient safety risk

Step 4: KPI List

- Total Patient Admissions
- Bed Occupancy Rate
- Average Length of Stay
- Department-wise Patient Load
- Readmission Rate
- Disease Frequency Trend
- Emergency Admission %
- Discharge Rate
- Age Group Distribution
- Mortality Rate (if applicable)

- Treatment Success Rate
- Data Error Rate

Step 5: Business Insight

Healthcare analytics improves hospital efficiency, patient safety, and resource allocation. Monitoring bed occupancy and disease trends supports capacity planning. Clean and accurate data is critical because incorrect health records can directly impact patient outcomes.

9 Real Estate Sales Dashboard

Domain: Real Estate

Step 1: Dataset Understanding

Columns:

- Property ID
- Location
- Price
- Size
- Status (Sold/Unsold)

Step 2: Cleaning Tasks

- Fix price formats
- Remove duplicate listings
- Categorize property types

Step 3: Business KPIs

- **Total Sales Revenue** → business growth
- **Sold vs Unsold Percentage** → inventory health
- **Average Property Price** → pricing benchmark
- **Location-wise Revenue** → demand analysis
- **Price per Sq. Ft.** → valuation indicator
- **Sales Conversion Rate** → deal closure efficiency
- **Time to Sell** → market speed indicator
- **Unsold Inventory Value** → blocked capital

- **Property Type Demand %** → buyer preference
- **Monthly Sales Trend** → investment planning metric

Step 4: KPI List

- **Total Sales Revenue**
- **Sold vs Unsold Percentage**
- **Average Property Price**
- **Price per Sq. Ft.**
- **Location-wise Revenue**
- **Sales Conversion Rate**
- **Time to Sell**
- **Unsold Inventory Value**
- **Property Type Demand %**
- **Monthly Sales Trend**
- **Revenue Growth Rate**
- **Inventory Turnover Rate**

Step 5: Business Insight

Real estate insights highlight high-demand locations, pricing trends, and unsold inventory risks. Tracking sales velocity and revenue growth helps investors allocate capital wisely. Data-driven decisions reduce financial risk in high-value property transactions.

10. Tourism & Hospitality

Domain: Travel & Hospitality

Step 1: Dataset Columns

- Hotel Name
- Room Type
- Booking Date
- Revenue

- Customer Type

Step 2: Cleaning Tasks

- Remove cancellations
- Standardize seasons
- Fix booking dates

Step 3: Business KPIs

- **Occupancy Rate** → hotel utilization
- **Revenue per Available Room (RevPAR)** → profitability metric
- **Average Daily Rate (ADR)** → pricing efficiency
- **Peak Season Revenue %** → seasonal demand strength
- **Cancellation Rate** → revenue leakage
- **Customer Segment Contribution** → target market analysis
- **Booking Lead Time** → demand forecasting
- **Repeat Guest Percentage** → customer loyalty
- **Room Type Revenue Contribution** → product performance
- **Monthly Revenue Trend** → financial planning

Step 4: KPI List

- Occupancy Rate
- Revenue per Available Room (RevPAR)
- Average Daily Rate (ADR)
- Cancellation Rate
- Peak Season Revenue %
- Booking Lead Time
- Repeat Guest Percentage
- Room Type Revenue Contribution
- Customer Segment Contribution
- Monthly Revenue Trend

- Seasonal Demand Trend
- Profit per Room

Step 5: Business Insight:

Hospitality analytics reveals seasonal demand patterns, occupancy trends, and revenue optimization opportunities. Monitoring cancellations and repeat guest rates improves pricing strategy and customer retention. Data enables hotels to maximize profitability during peak seasons and minimize losses during low demand periods.